

High Performance Computing with GPUs
Complex Computing Problem
Deliverable 1

Submitted By:

Masooma Ali 23I-0623

Huma Taj 23I-0743

Performance Profiling & GPU Optimization Report

KLT Feature Tracker Acceleration on GPUs

1. Objective

The goal of this profiling session was to analyze the program's runtime performance, identify CPU bottlenecks, and determine which parts can be accelerated using GPU-based computation.

2. Tools & Setup

Tools used include gprof (for profiling), gprof2dot (for visualization conversion), and Graphviz (to render the call graph). The commands used are listed below:

Run Makefile: make

Execution: ./example3

Profiling Output: gprof ./example3 gmon.out > profile.txt

Visualization: gprof2dot -f prof profile.txt | dot -Tpng -o profile.png

3. Profiling Results Summary

The profiling revealed that most of the runtime is spent in convolution functions. The flat profile shows the time spent per function, while the call graph identifies function relationships.

4. GPU Optimization Recommendations

The profiling revealed that convolution-based operations are the main CPU bottleneck. The convolution functions (_convolveSeparate, _convolveImageVert, _convolveImageHoriz) consume ~60% of total runtime. By offloading these to the GPU, significant performance improvements can be achieved.

5. Function Performance Summary

Function	% of Total Time	Called By	Calls
<code>_convolveSeparate()</code>	60	KLTTrackFeatures	<code>_convolveImageVert()</code> , <code>_convolveImageHoriz()</code>
<code>_convolveImageVert()</code>	36	<code>_convolveSeparate()</code>	-
<code>_convolveImageHoriz()</code>	24	<code>_convolveSeparate()</code>	-
<code>_computeGradientSum()</code>	10	<code>_trackFeatures()</code>	-
<code>_interpolate()</code>	6	<code>_trackFeatures()</code>	-
<code>_trackFeatures()</code>	5	<code>main()</code>	-

